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Special Article

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Recommendations for Influenza Vaccination in Immunocompromised Patients

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ABSTRACT

Annual influenza vaccination is strongly recommended for immunocompromised individuals, who are at a higher risk of severe infection. The Korean Society of Infectious Diseases has revised its influenza vaccination recommendations for solid organ transplant recipients, advocating for the use of highly immunogenic vaccines, such as high-dose inactivated influenza vaccines or MF59-adjuvanted inactivated influenza vaccines, to enhance vaccine efficacy. While further research is needed to extend these recommendations to other immunocompromised populations, these vaccines have demonstrated improved immunogenicity without compromising safety. Expanding vaccination guidelines to additional immunocompromised groups should be considered as more evidence emerges.

Keywords: Influenza; Vaccination; Immunocompromised host

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SUMMARY OF RECOMMENDATIONS

1. Immunocompromised individuals should receive the influenza vaccine recommended annually based on their age.
2. Solid organ transplant patients aged 18–64 years who are taking immunosuppressive agents may receive a high-immunogenicity influenza vaccine (high-dose vaccine or MF59-adjuvanted vaccine).

Immunocompromised individuals face an increased risk of severe influenza infection, hospitalization, and mortality. Therefore, as a preventative measure, annual influenza vaccination before the start of flu season is recommended [1, 2]. However, the inhaled live attenuated influenza nasal spray vaccine (Flumist®, MedImmune, LLC, Gaithersburg, MD, USA) is contraindicated in immunocompromised individuals due to safety concerns. A meta-analysis has demonstrated that influenza vaccines provide significant protective effects even in immunocompromised individuals. However, the degree of immunosuppression and the immunogenicity of the vaccine vary considerably depending on the underlying immunocompromising condition [3-5]. Solid organ transplant recipients exhibit lower vaccine induced immunogenicity following influenza vaccination compared to healthy adults, and a considerable number of solid organ transplant recipients still contract influenza despite vaccination [6, 7]. This highlights the need for highly immunogenic influenza vaccines, such as high-dose inactivated influenza vaccines (HD-IIV, Efluelda®, Sanofi Pasteur, Lyon, Auvergne-Rhône-Alpes, France) and MF59-adjuvanted inactivated influenza vaccines (MF59-aIIV, Fluad®, Seqirus, Inc., Holly Springs, NC, USA), to enhance immune responses in solid organ transplant recipients [8].

The global population of solid organ transplant recipients is increasing, with improved post-transplant survival rates [9]. In Korea, the average number of solid organ transplants over the past five years has been 4,096 cases annually. The 10-year survival rates of transplant recipients range from 77.4–80.1%, with kidney transplants (85.9–88.8%), pancreas transplants (85.5–88.0%), liver transplants (79.3–83.0%), heart transplants (60.9–65.1%), and lung transplants (28.5–35.9%), presented in decreasing order [10]. In solid organ transplant recipients, influenza virus infection commonly causes complications such as pneumonia (21–49%) and acts as a risk factor for transplant organ dysfunction, intensive care unit admission, and mortality [7, 11].

A meta-analysis of seven studies conducted by the Advisory Committee on Immunization Practices in the United States found that while HD-IIV (2–4 times the standard dose) or MF59-aIIV did not significantly reduce influenza-related hospitalization risks, they exhibited improved immunogenicity compared to standard-dose influenza vaccines in solid organ transplant recipients [12]. Specifically, HD-IIV demonstrated 1.03–1.22 times higher seroprotection rates and 1.67–2.46 times higher seroconversion rates compared to standard-dose vaccines, while MF59-aIIV exhibited 1.06–1.20 times higher seroprotection rates and 1.37–1.64 times higher seroconversion rates [13]. In the STOP-FLU trial, which directly compared the immunogenicity of HD-IIV, MF59-aIIV, and standard-dose influenza vaccines in solid organ transplant recipients, the proportion of individuals exhibiting a fourfold or greater increase in antibody titers against at least one influenza subtype was 66% for HD-IIV, 60% for MF59-aIIV, and 42% for standard-dose vaccines. These findings confirm the relatively higher immunogenicity of high-dose and adjuvanted vaccines [14]. Moreover, compared to standard-dose influenza vaccines, highly immunogenic influenza vaccines (HD-IIV or MF59-aIIV) did not increase the risks of transplant rejection, neuroinflammatory adverse reactions, or autoimmune complications in solid organ transplant recipients [12-14]. Both vaccine types demonstrated good tolerability. Based on the high immunogenicity and safety profiles of HD-IIV and MF59-aIIV, solid organ transplant recipients aged 18–64 who are taking immunosuppressive therapy may receive highly immunogenic influenza vaccines.

Immunocompromising conditions encompass a wide range of diseases, with significant differences in immunosuppression severity and infection risk. Due to insufficient evidence on influenza burden and vaccine efficacy across all immunocompromised populations, a uniform recommendation for highly immunogenic influenza vaccines cannot be made at this time. Therefore, we primarily recommend highly immunogenic influenza vaccines for solid organ transplant recipients, who require lifelong immunosuppression, face a high disease burden from severe influenza, and have available comparative immunogenicity data. As additional studies emerge, vaccination recommendations may be expanded to other immunocompromised populations.

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Conflict of Interest
JYS and WSC serve on the Immunization Committee of the Korea Disease Control and Prevention Agency (KDCA). WBP is an editor in chief of Infect Chemother; JYS and JYC are editorial board of Infect Chemother; however, they did not involve in the peer reviewer selection, evaluation, and decision process of this article. Otherwise, no potential conflicts of interest relevant to this article was reported.

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Conceptualization: JYS, WSC. Data curation: JYS. Writing – original draft: JYS. Writing – review & editing: JYS, KTK, WBP, JYN, SHP, EJC, MJC, JYC, JYH, WSC.

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